Numerical Methods II

If a data consisting of $(n+1)$ ordered pairs (x_i, y_i), $i = 0, 1, 2, \cdots, n$ of an unknown function 'y' depending on 'x' is available, (this data could be experimental or observations) estimating value of function employing polynomials, obtain the derivatives, determining an approximate value of definite integral would be taken up in this chapter.

Finite differences:

Let $x = x_0, x_1, x_2, \cdots, x_n$ be equally spaced arguments, $y = f(x) = y_0, y_1, y_2, \cdots, y_n$ be the corresponding ordinates of a function, $f(x)$ is not available explicitly. Let $h = (x_1 - x_0), (x_2 - x_1), \cdots, (x_n - x_{n-1})$ known as interval of differencing.

Define first order forward difference of ordinate (function) as:

$$\boxed{\Delta(y_r) = (y_{r+1} - y_r)}, \quad r = 0, 1, 2, \cdots, n$$

'Δ' is known as 'forward difference' operator.

$\Delta y_0 = (y_1 - y_0)$, $\Delta y_1 = (y_2 - y_1)$, $\Delta y_2 = (y_3 - y_2)$ and so on

Second forward difference is defined as:

$$\Delta^2(y_r) = \Delta(\Delta y_r)$$

$$\Delta^2 y_r = \Delta(y_{r+1} - y_r)$$

$$\Delta^2 y_r = \Delta(y_{r+1}) - \Delta(y_r)$$

$$\therefore \Delta^2 y_r = [(y_{r+2} - y_{r+1}) - (y_{r+1} - y_r)]$$

$$\Delta^2 y_r = (y_{r+2} - 2y_{r+1} + y_r)$$

Higher order forward differences are defined in a similar manner

also, $\Delta f(x) = [f(x+h) - f(x)]$

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The backward difference of ordinates are defined as :

$$\boxed{\nabla(y_r) = (y_r - y_{r-1})}, \quad r = 1, 2, 3, \ldots n$$

here ∇ is known as backward difference operator

$\nabla y_1 = (y_1 - y_0)$, $\nabla y_2 = (y_2 - y_1)$, $\nabla y_3 = (y_3 - y_2)$, ...

Second backward difference is defined as:

$$\nabla^2(y_r) = \nabla(\nabla y_r)$$

$$\nabla^2 y_r = \nabla(y_r - y_{r-1})$$

$$\nabla^2 y_r = \nabla(y_r) - \nabla(y_{r-1})$$

$$\nabla^2 y_r = [(y_r - y_{r-1}) - (y_{r-1} - y_{r-2})]$$

$$\therefore \nabla^2 y_r = [(y_r - 2y_{r-1} + y_{r-2})]$$

Higher order backward differences are defined in similar manner.

also, $\nabla f(x) = [f(x) - f(x-h)]$

It should be noted that forward differences of ordinate and backward differences of ordinates are related.

These could be tabulated in form of a table called as finite difference table.

Observe that $\nabla y_1 = (y_1 - y_0) = \Delta(y_0)$

$\nabla(y_2) = \Delta(y_1)$ and so on

$\nabla^2(y_2) = \Delta^2(y_0)$ (verify this with definition)

If $y = c$, a constant

then $\Delta(y) = \Delta(c) = 0$ as constant value does not change.

also, $\Delta[c f(x)] = c \Delta f(x)$, c is a constants

$\Delta[c_1 f(x) \pm c_2 g(x)] = c_1 \Delta f(x) \pm c_2 \Delta g(x)$

where c_1 and c_2 are constants, $\nabla^r y_n = \Delta^r y_{n-r}$

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If $y = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n$ $(a_0 \neq 0)$ is a polynomial of degree n, then it can be proved that $\Delta^n(y) = (a_0 n! h^n)$ (a constant)

and $\Delta^{n+1}(y) =$ zero

Finite difference table

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
x_0	y_0				
		Δy_0			
x_1	y_1		$\Delta^2 y_0$		
		Δy_1		$\Delta^3 y_0$	
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$
		Δy_2		$\Delta^3 y_1$	
x_3	y_3		$\Delta^2 y_2$		
		Δy_3			
x_4	y_4				

Δy_0, $\Delta^2 y_0$, $\Delta^3 y_0$, $\Delta^4 y_0$ are known as leading differences, as forward and backward differences are related same table is used for backward differences also, but with different interpretation.

Examples:

1. Evaluate $\Delta(\tan^{-1} x)$
2. Evaluate $\Delta^6[(2-3x)(3-4x^2)(2+5x^3)]$, when $h = 1$
3. If $f(x) = e^{ax}$, find $\Delta^n f(x)$, n is positive integer.

Sol: (1) Let $f(x) = \tan^{-1}(x)$

$\therefore f(x+h) = \tan^{-1}(x+h)$

by definition $\Delta f(x) = f(x+h) - f(x)$

$\therefore \Delta \tan^{-1}(x) = \tan^{-1}(x+h) - \tan^{-1}(x)$

$$= \tan^{-1}\left[\frac{(x+h)-x}{1+(x+h)x}\right]$$

$$\therefore \Delta \tan^{-1}(x) = \tan^{-1}\left(\frac{h}{1+hx+x^2}\right)$$

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Sol (2): Let $f(x) = (2-3x)(3-4x^2)(2+5x^3)$

here $f(x)$ is a polynomial of degree 6 with coefficient of x^6 being: $(-3)(-4)(5) = 60 = a_0$ (say)

$\therefore \quad \Delta^6 f(x) = a_0\, n!\, h^n$ (Standard result)

$= (60)\, 6!\, 1^6 \quad (\because h = 1$ given)

$= 43200$

Sol (3): Let $f(x) = e^{ax}$

$\therefore \quad f(x+h) = e^{a(x+h)}$

$\therefore \quad \Delta f(x) = [f(x+h) - f(x)] = e^{(ax+ah)} - e^{ax}$

$= e^{ax} \cdot e^{ah} - e^{ax} = e^{ax}(e^{ah} - 1)$

$\Delta^2 f(x) = \Delta\{\Delta f(x)\}$

$= \Delta\{(e^{ah} - 1)\, e^{ax}\}$

$= (e^{ah} - 1)\, \Delta(e^{ax}) = (e^{ah} - 1)[(e^{ah} - 1)\, e^{ax}]$

$= (e^{ah} - 1)^2 e^{ax}$

hence, $\Delta^n f(x) = (e^{ah} - 1)^n e^{ax}$

4. If $x = 0, 1, 2, 3, 4$ and corresponding values of a function $y = 10, 20, 30, 40, 50$ respectively, construct a forward difference table for above data.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	10				
		10			
1	20		0		
		10		0	
2	30		0		0
		10		0	
3	40		0		
		10			
4	50				

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Interpolation :

Interpolation is the 'art of reading between the lines in a table'. Interpolation is an estimation of a value within two known values in a sequence of values. Extrapolation is an estimation of a value based on extending a known sequence of values.

It is necessary because in scienc and engineering we often need to deal with discrete experimental data. The given data is approximated with polynomials passing through points and functional values can be estimated.

Newton - Gregory forward interpolation formula :

Let $x = x_0, x_1, x_2, \ldots, x_n$ be equally spaced arguments, $y = y_0, y_1, y_2, \ldots, y_n$ be corresponding ordinates of $y = f(x)$, Let $x = x_0 + rh$, then the value of y can be estimated using,

$$y = y_0 + \frac{r}{1!}(\Delta y_0) + \frac{r(r-1)}{2!}(\Delta^2 y_0) + \frac{r(r-1)(r-2)}{3!}(\Delta^3 y_0) + \cdots$$

where $r = \frac{(x - x_0)}{h}$, the above expression is a polynomial, used to determine y for x near the beginning of tabulated values.

The above expression is called as Newton's forward interpolation formula (or Newton - Gregory forward interpolation formula).

Here 'h', the interval of differencing is a constant

Newton - Gregory backward interpolation formula :

Let (x_i, y_i), $i = 0, 1, 2, \ldots, n$ be $(n+1)$ points here $h = x_{i+1} - x_i$, a constant, $y = f(x)$ and $x = x_n + ph$, then value of y can be estimated using

$$y = y_n + \frac{p}{1!}(\nabla y_n) + \frac{p(p+1)}{2!}(\nabla^2 y_n) + \frac{p(p+1)(p+2)}{3!}(\nabla^3 y_n) + \cdots$$

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where $p = \frac{(x - x_n)}{h}$, this preceding expression is called as Newton (or Newton - Gregory) backward interpolation formula, used to estimate y near the end of the tabulated values.

Examples:

1. Find a polynomial that passes through the points (3, 1), (5, 3), (7, 8), (9, 16) and (11, 27)

Sol: Here

	x_0	x_1	x_2	x_3	x_4
x :	3	5	7	9	11
y :	1	3	8	16	27

Consider the finite difference table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
3	1				
		[2]			
5	3		[3]		
		5		[0]	
7	8		3		[0]
		8		0	
9	16		3		
		11			
11	27				

Here $\Delta y_0 = 2$, $\Delta^2 y_0 = 3$, $\Delta^3 y_0 = 0$, $\Delta^4 y_0 = 0$

$x_0 = 3$, $h = 2$, $y_0 = 1$

Since $x = x_0 + rh = 3 + r(2)$

$\therefore r = \frac{(x-3)}{2}$

By Newton's forward interpolation formula,

$$y = y_0 + \frac{r}{1!}(\Delta y_0) + \frac{r(r-1)}{2!}(\Delta^2 y_0) + \frac{r(r-1)(r-2)}{3!}(\Delta^3 y_0) + \cdots$$

$$\therefore y = 1 + \frac{(x-3)}{2}(2) + \frac{(x-3)\left(\frac{x-3}{2} - 1\right)}{2!}(3) + 0$$

$$y = 1 + (x-3) + \frac{3}{8}(x-3)(x-5)$$

$$y = \frac{3}{8}x^2 - 2x + \frac{29}{8}$$

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2. The following table gives the monthly income of 2000 employees in a company. Using Newton's forward interpolation technique, estimate the number of persons who have monthly income between Rs 8000 and Rs 10000.

Monthly Income (Rs)	Below 5000	5000 – 10000	10000 – 15000	15000 – 20000	20000 – 25000
Number of employees	535	660	470	270	65

Sol: We shall form cumulative frequency distribution

Let x be upper limit of income and

y be total number of person, hence

given data is:

x :	5000	10000	15000	20000	25000
y :	535	1195	1665	1935	2000

Consider the finite difference table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
5000	535				
		660			
10000	1195		-190		
		470		-10	
15000	1665		-200		5
		270		-5	
20000	1935		-205		
		65			
25000	2000				

Here $x_0 = 5000$, $y_0 = 535$, from differe table

$\Delta y_0 = 660$, $\Delta^2 y_0 = -190$, $\Delta^3 y_0 = -10$, $\Delta^4 y_0 = 5$

We shall determine y at $x = 8000$

Since $x = x_0 + rh$, $8000 = 5000 + r(5000)$

$\therefore r = (8000 - 5000)/5000 = 0.6$

$$y = y_0 + r(\Delta y_0) + \frac{r(r-1)}{2!}(\Delta^2 y_0) + \cdots$$

$$y = 535 + (0.6)(660) + \frac{(0.6)(0.6-1)}{2!}(-190) + \frac{(0.6)(0.6-1)(0.6-2)}{3!}(-10) + \frac{(0.6)(0.6-1)(0.6-2)(0.6-3)}{4!}(5) + \cdots$$

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$$y = 953.072 \approx 953$$

Hence, no. of employee whose income is between Rs 8000 and Rs10000 is

$$y(10000) - y(8000) = 1195 - 953 = 242$$

3. Using Newton's backward interpolation technique, fit a polynomial of degree three which takes the following values.

x :	3	4	5	6
y :	6	24	60	120

Sol: Consider the finite difference table:

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
3	6			
		18		
4	24		18	
		36		6
5	60		24	
		60		
6	120			

Here $x_n = x_3 = 6$, $y_3 = 120$, $\nabla y_3 = 60$, $\nabla^2 y_3 = 24$, $\nabla^3 y_3 = 6$

By Newton's backward interpolation formula,

$$y = y_3 + \frac{p}{1!}(\nabla y_3) + \frac{p(p+1)}{2!}(\nabla^2 y_3) + \frac{p(p+1)(p+2)}{3!}(\nabla^3 y_3) + \cdots$$

where $p = \frac{x - x_n}{h} = \frac{x - x_3}{h} = \frac{(x-6)}{1} = (x-6)$

$$\therefore \quad y = 120 + (x-6)(60) + \frac{(x-6)(x-5)}{2!}(24) + \frac{(x-6)(x-5)(x-4)}{3!}(6) + \cdots$$

$$\therefore \quad y = x^3 - 3x^2 + 2x$$

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4. The area A of a circle for different diameters d is given in the following table:

d :	80	85	90	95	100
A :	5026	5674	6362	7088	7854

Using relevant interpolation technique, calculate the area of a circle with diameter 105 units.

Sol: Here, we have an example of extrapolation, further $d = 105$ is in the later part of tabulated values, hence Newton's backward interpolation is preferred; $A = f(d)$, in the given geometrical consideration.

Consider the finite difference table:

d	A	∇A	$\nabla^2 A$	$\nabla^3 A$	$\nabla^4 A$
80	5026				
		648			
85	5674		40		
		688		−2	
90	6362		38		4
		726		2	
95	7088		40		
		766			
100	7854				

$d_n = d_4 = 100$, $A_4 = 7854$, $\nabla A_4 = 766$, $\nabla^2 A_4 = 40$, $\nabla^3 A_4 = 2$, $\nabla^4 A_4 = 4$, $h = 5$

We need A at $d = 105$, $\quad d = d_4 + ph$

$105 = 100 + p(5) \quad \therefore p = \frac{(105 - 100)}{5} = 1$

By Newton's backward interpolation,

$$A(105) = A_4 + \frac{p}{1!}(\nabla A_4) + \frac{p(p+1)}{2!}(\nabla^2 A_4) + \cdots$$

$$A = 7854 + 1(766) + \frac{1.2}{2!}(40) + \frac{1.2.3}{3!}(2) + \frac{1.2.3.4}{4!}(4) + \cdots$$

$$A = 8666$$

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Practice Examples:

1. The following table gives the distances (in miles) of the visible horizon for the given heights (in feet) above the earth's surface

x :	200	250	300	350	400
y :	15.04	16.81	18.42	19.9	21.27

Find the value of y at $x = 218$ ft and $x = 389$ ft.

(15.6772 , 20.9686)

2. The deflection d measured at various distances x from one end of a cantilever are given below. Find d when $x = 0.95$

x :	0.0	0.2	0.4	0.6	0.8	1.0
y :	0.0000	0.0350	0.1170	0.2165	0.2995	0.334

($d = 0.3314$ at $x = 0.95$)

3. The marks secured by 280 candidates in an examination are given below. Using Newton's forward interpolation technique estimate the number of candidates who secured marks between 45 and 50.

Marks :	Below 30	30-40	40-50	50-60	60-70	70-80
Number of candidates :	35	49	62	74	40	20

$y(45) = 111.15 \approx 111$, $y(50) = 146$ $\therefore$ $146 - 111 = 35$ candidates

4. Find the interpolating polynomial $P(x)$ fitting the data given below, hence find $P(0.5)$

x :	-2	-1	0	1	2	3
$P(x)$:	-3	3	3	3	9	27

$P(x) = x^3 - x + 3$

$P(0.5) = 2.625$

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Lagrange's interpolation:

Newtons forward and backward interpolation techniques to estimate the values of function are not applicable if the arguments are unequally spaced. Also, it is not possible to obtain inverse i.e. if $y = f(x)$ it is not possible to determine x for given y. Lagrange interpolation has advantage over the other two methods discussed earlier.

Let (x_i, y_i), $i = 0, 1, 2, \ldots, n$ be given data of a function $y = f(x)$ such that $x_0, x_1, \ldots, x_n$ are not necessarily spaced equally; then Lagrange polynomial passing through these points is given by:

$$P(x) = \frac{(x-x_1)(x-x_2)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)\cdots(x_1-x_n)} y_1$$
$$+ \cdots + \frac{(x-x_0)(x-x_1)\cdots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\cdots(x_n-x_{n-1})} y_n$$

In the above expression it is possible to interchange x and y (i.e. $y = f(x) = P(x)$), using the resulting expression we can obtain x for given y called as inverse Lagrange's interpolation.

$$x = \frac{(y-y_1)(y-y_2)\cdots(y-y_n)}{(y_0-y_1)(y_0-y_2)\cdots(y_0-y_n)} x_0 + \frac{(y-y_0)(y-y_2)\cdots(y-y_n)}{(y_1-y_0)(y_1-y_2)\cdots(y_1-y_n)} x_1$$
$$+ \cdots + \frac{(y-y_0)(y-y_1)\cdots(y-y_{n-1})}{(y_n-y_0)(y_n-y_1)\cdots(y_n-y_{n-1})} x_n$$

Examples:

1. Using Lagrange's interpolation technique determine a polynomial which passes through the points $y(1) = 0$, $y(3) = 12$, $y(4) = 33$ and $y(6) = 135$.

Sol: Here $x_0 = 1$, $x_1 = 3$, $x_2 = 4$, $x_3 = 6$
$y_0 = 0$, $y_1 = 12$, $y_2 = 33$, $y_3 = 135$
(data)

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$$\therefore y = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 +$$
$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3 \quad (1)$$

$$\therefore y = \frac{(x-3)(x-4)(x-6)}{(1-3)(1-4)(1-6)}(0) + \frac{(x-1)(x-4)(x-6)}{(3-1)(3-4)(3-6)}(12) +$$
$$\frac{(x-1)(x-3)(x-6)}{(4-1)(4-3)(4-6)}(33) + \frac{(x-1)(x-3)(x-4)}{(6-1)(6-3)(6-4)}(135)$$
$$= 0 + \frac{(x^3-11x^2+34x-12)}{2(-1)(-3)}(12) + \frac{(x^3-10x^2+27x-18)}{3(1)(-2)}(33)$$
$$+ \frac{(x^3-8x^2+19x-12)}{(5)(3)(2)}(135)$$

$$\therefore y = [redacted] (x^3-3x^2+5x-3)$$

2. The following table gives the viscosity of an oil as function of temperature, find the viscosity of oil at temperature of 140°c.

Temperature (°c) :	110	130	160	190
Viscosity :	10.8	8.1	5.5	4.8

Sol: Here $y = f(x)$, where x is temperature and y is the viscosity of oil.

$x_0 = 110$, $x_1 = 130$, $x_2 = 160$, $x_3 = 190$

$y_0 = 10.8$, $y_1 = 8.1$, $y_2 = 5.5$ $y_3 = 4.8$

We need to determine y at $x = 140$, using the above eqn (1), substituting the values for x and y we have:

$$y = \frac{(140-130)(140-160)(140-190)}{(110-130)(110-160)(110-190)}(10.8) +$$
$$\frac{(140-110)(140-160)(140-190)}{(130-110)(130-160)(130-190)}(8.1) +$$

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$$+ \frac{(140-110)(140-130)(140-190)}{(160-110)(160-130)(160-190)}(5.5) +$$

$$\frac{(140-110)(140-130)(140-160)}{(190-110)(190-130)(190-160)}(4.8)$$

$\therefore \quad y(140) \approx 7.0333$

3. Find the approximate root of $f(x)=0$, given that $f(5)=-30$, $f(8)=-12$, $f(11)=4$ and $f(14)=20$

Sol: Let $y=f(x)$ and given data is:

$x_0=5$, $x_1=8$, $x_2=11$, $x_3=14$ and $y_0=-30$, $y_1=-12$, $y_2=4$ and $y_3=20$

Now it required to determine 'x' such that $f(x)=0$ i.e. $y=0$, by inverse Lagrange's interpolation,

$$x = \frac{(y-y_1)(y-y_2)(y-y_3)}{(y_0-y_1)(y_0-y_2)(y_0-y_3)}x_0 + \frac{(y-y_0)(y-y_2)(y-y_3)}{(y_1-y_0)(y_1-y_2)(y_1-y_3)}x_1 +$$

$$\frac{(y-y_0)(y-y_1)(y-y_3)}{(y_2-y_0)(y_2-y_1)(y_2-y_3)}x_2 + \frac{(y-y_0)(y-y_1)(y-y_2)}{(y_3-y_0)(y_3-y_1)(y_3-y_2)}x_3$$

Substituting the data available and setting $y=0$ in above expression, we get:

$$x = -\frac{(12)(4)(20)}{(18)(34)(50)}(5) + \frac{(30)(4)(20)}{(18)(16)(32)}(8) +$$

$$\frac{(30)(12)(20)}{(34)(16)(16)}(11) + \frac{(-30)(12)(4)}{(50)(32)(16)}(14)$$

$\therefore \quad x \approx 10.2382$

Hence, approximate root of $f(x)=0$ is $x=10.2382$

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Practice Examples:

1. Using Lagrange's interpolation find a polynomial $L(x)$ that passes through the points $(2, 30)$, $(4, 36)$, $(5, 60)$ and $(7, 120)$.

$(L(x) = -x^3 + 18x^2 - 77x + 120)$

2. Determine $f(2)$ using Lagrange's interpolation technique for the following data:

x :	-1	0	1	3
$f(x)$:	4	13	40	96

$(f(2) = 72)$

3. Given:

x :	12	20	25	30
y :	13.6	5.8	3.4	2.0

by using inverse interpolation find the value of x when $y = 12$.

$(y \approx 14.73)$

Numerical differentiation:

If a data consisting of $(n+1)$ ordered pairs of an an unknown function $y = f(x)$ is given, the process of finding the derivatives of function at a specified point is called 'numerical differentiation.

The process involves approximating data by interpolating polynomial and obtain its derivative.

We shall restrict our discussion on finding derivatives to:

Newton's forward and Newton's backward interpolation polynomials.

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If $D = \frac{d}{dx}$, Δ and ∇ are forward and backward operators respectively it can be proved that:

$$D = \frac{1}{h}\{\log(1+\Delta)\}, \quad D^2 = \frac{1}{h^2}\{\log(1+\Delta)\}^2$$

and $$D = -\frac{1}{h}\{\log(1-\nabla)\}, \quad D^2 = \frac{1}{h^2}\{\log(1-\nabla)\}^2$$

Taylor series expansion for $\log(1+x)$ and $-\log(1-x)$ at origin is

$$\log(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + - \cdots$$

$$-\log(1-x) = x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \frac{1}{4}x^4 + \cdots$$

$$\therefore y'(x_0) = D(y)_{x=x_0} = \frac{1}{h}\left[\Delta - \frac{1}{2}\Delta^2 + \frac{1}{3}\Delta^3 - \frac{1}{4}\Delta^4 + \cdots\right]y_0$$

$$y'(x_0) = \frac{1}{h}\left[(\Delta y_0) - \frac{1}{2}(\Delta^2 y_0) + \frac{1}{3}(\Delta^3 y_0) - + \cdots\right]$$

Similarly, $$y''(x_0) = \frac{1}{h^2}\left[(\Delta^2 y_0) - (\Delta^3 y_0) + \frac{11}{12}(\Delta^4 y_0) - + \cdots\right]$$

$$y'(x_n) = \frac{1}{h}\left[(\nabla y_n) + \frac{1}{2}(\nabla y_n)^2 + \frac{1}{3}(\nabla y_n)^3 + \cdots\right]$$

$$y''(x_n) = \frac{1}{h^2}\left[(\nabla^2 y_n) + (\nabla^3 y_n) + \frac{11}{12}(\nabla^4 y_n) + \cdots\right]$$

Alternatively, a polynomial through given points can be obtained, from the resulting polynomial $f'(x)$, $f''(x)$ can be obtained at given point.

Examples:

1. A rod is rotating in a plane, the following data gives the angle θ (in radians) turned by the rod, recorded for different values of time 't' (in sec).

t	0	0.2	0.4	0.6	0.8	1.0	1.2
θ	0	0.12	0.49	1.12	2.02	3.2	4.67

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Find the angular velocity and angular acceleration of the rod when $t = 0.2$ sec.

Sol: Here $\theta = f(t)$, 't' is equally spaced with $h = 0.2$

We need to determine $\left(\frac{d\theta}{dt}\right)$ and $\left(\frac{d^2\theta}{dt^2}\right)$ at $t = 0.2$

this point is in beginning of tabulated values.

Consider the finite difference table:

t	θ	$\Delta\theta$	$\Delta^2\theta$	$\Delta^3\theta$	$\Delta^4\theta$
0	0				
		0.12			
0.2	0.12		0.25		
		0.37		0.01	
0.4	0.49		0.26		0
		0.63		0.01	
0.6	1.12		0.27		0
		0.90		0.01	
0.8	2.02		0.28		0
		1.18		0.01	
1.0	3.2		0.29		
		1.47			
1.2	4.67				

Identifying the forward differences associated with $t = 0.2$,

$$\frac{d\theta}{dt} = \frac{1}{h}\left[\Delta\theta_0 - \frac{1}{2}\Delta^2\theta_0 + \frac{1}{3}\Delta^3\theta_0 - + \cdots\right]$$

$$= \frac{1}{(0.2)}\left[0.37 - \frac{(0.26)}{2} + \frac{(0.01)}{3}\right]$$

$$= 1.2167 \text{ rad/sec}$$

$$\frac{d^2\theta}{dt^2} = \frac{1}{h^2}\left[\Delta^2\theta_0 - \Delta^3\theta_0 + \frac{11}{12}\Delta^4\theta_0 - + \cdots\right]$$

$$= \frac{1}{(0.04)}\left[0.26 - 0.01 + 0\right]$$

$$= 6.25 \text{ rad/sec}^2.$$

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2. From the following table, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.25$

x	1.0	1.05	1.1	1.15	1.2	1.25	1.3
y	1.00	1.0247	1.04889	1.07256	1.0957	1.1183	1.14035

Sol: Consider the finite difference table:

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1	1				
		0.0247			
1.05	1.0247		−0.00051		
		0.02419		−0.00001	
1.10	1.04889		−0.00052		0
		0.02367		−0.00001	
1.15	1.07256		−0.00053		0
		0.02314		−0.00001	
1.2	1.0957		−0.00054	−0.00001	0
		0.0226			
1.25	1.1183		−0.00055		
		0.02205			
1.30	1.14035				

we have $y' = \frac{1}{h}\left[\nabla y_n + \frac{\nabla^2 y_n}{2} + \frac{\nabla^3 y_n}{3} + \cdots\right]$

$\therefore y'(1.25) = \frac{1}{0.05}\left\{0.0226 + \frac{1}{2}(-0.00054) + \frac{1}{3}(-0.00001)\right\}$

$y'(1.25) = 0.44653$

again, $y'' = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \cdots\right]$

$y''(1.25) = \frac{1}{(0.05)^2}\left[-0.00054 - 0.00001\right]$

$y''(1.25) = -0.22$

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Numerical Integration:

In many applications, obtaining exact solution of a definite integral may not possible. The process of obtaining value of a definite integral, when function is specified in form of tabulated values is known as numerical integration. This method applied for a function of single variable is known as 'Quadrature'.

Newton-Cotes Quadrature:

Consider $I = \int_a^b f(x)\,dx \quad (1)$

Divide the interval of integration $[a, b]$ into n equal parts consisting of $(n+1)$ ordinates.

Let $a = x_0$ and $b = x_n = x_0 + nh$, $h = \frac{(b-a)}{n}$

If $f(x)$ is approximated by Newton forward interpolation polynomial i.e.

$$y = f(x) = y_0 + \frac{r}{1!}(\Delta y_0) + \frac{r(r-1)}{2!}(\Delta^2 y_0) + \cdots$$

where $x = x_0 + rh$, $dx = 0 + h\,dr$

when $x = x_0$, $r = 0$, $x = x_n = x_0 + nh = x_0 + rh$

$r = n$

(1) becomes,

$$I = \int_{r=0}^{n} h\left[y_0 + \frac{r}{1!}(\Delta y_0) + \frac{(r^2 - r)}{2!}(\Delta^2 y_0) + \cdots \right] dr$$

$$\therefore \quad I = h\left[y_0 r + \frac{\Delta y_0}{1!} r^2 + \frac{(\Delta^2 y_0)}{2!}\left(\frac{r^3}{3} - \frac{r^2}{2}\right) + \cdots \right]_{r=0}^{n}$$

$$\therefore \int_a^b f(x)\,dx = h\left\{ n y_0 + n^2(\Delta y_0) + \left(\frac{n^3}{3} - \frac{n^2}{2}\right)\frac{(\Delta^2 y_0)}{2} + \cdots \right\}$$

The above expression is known as 'Newton-Cotes' quadrature rule.

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If $f(x)$ is approximated by polynomial of degree two, three and six respectively in 'Newton-Cotes quadrature rule respectively, gives:

Simpson's $\frac{1}{3}$rd rule

Simpson's $\frac{3}{8}$th rule

Weddle's rule

The proof of these are beyond the scope of course, hence not included.

Simpson's $\frac{1}{3}$rd rule:

$$I = \int_a^b f(x)dx = \frac{h}{3}\left[(y_0+y_n) + 4(y_1+y_3+y_5+\cdots) + 2(y_2+y_4+y_6+\cdots)\right]$$

here $y = f(x)$, $[a,b]$ is divided into n equal parts (n is even) and $f(x)$ is polynomial of degree 2.

Simpson's $\frac{3}{8}$th rule:

$$I = \int_a^b f(x)dx = \frac{3h}{8}\left[(y_0+y_n) + 3(y_1+y_2+y_4+y_5+\cdots) + 2(y_3+y_6+\cdots)\right]$$

here $y = f(x)$, $[a,b]$ is divided into n equal parts (n is multiple of three) and $f(x)$ is a polynomial of degree 3.

Weddle's rule:

$$I = \int_{x_0}^{x_6} f(x)dx = \frac{3h}{10}\left[y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6\right] \quad (n=6)$$

$$I = \int_{x_0}^{x_{12}} f(x)dx = \frac{3h}{10}\left[(y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6) + (y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + y_{12})\right] \quad (n=12)$$

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Examples:

1. A curve is described by the points given below, using Simpson's rule, find the area bounded by the curve, x-axis and the ordinates $x=0$ and $x=4$.

x :	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y :	4.6	3.8	2.8	2.2	2.5	3.2	3.8	4.0	4.0

Sol: Here $A = \int_{x_0=0}^{x_8=4} f(x)\,dx$, $h = 0.5$, $n = 8$, nine ordinates are available, hence by Simpson's $\frac{1}{3}$rd rule

$$A = \frac{h}{3}[(y_0+y_8) + 4(y_1+y_3+y_5+y_7) + 2(y_2+y_4+y_6)]$$

$$= \frac{0.5}{3}[(4.6+4) + 4(3.8+2.2+3.2+4) + 2(2.8+2.5+3.8)]$$

$$= 13.2667$$

Note that in above example $n=8$ (even) hence only Simpson's $\frac{1}{3}$rd rule is applicable. 'n' corresponds to sub divisions of interval of integration, here $[0, 4]$.

2. The velocity of a train which starts from rest from a station and reaches the next ~~st~~ station in 12 minutes is recorded at interval of 2 minutes as follows:

t (hr)	0	2/60	4/60	6/60	8/60	$\frac{10}{60}$	$\frac{12}{60}$
v (kmph)	0	20	36	60	64	42	0

Find the distance between the stations using Simpson's $\frac{3}{8}$th rule

Sol: Here $h = 2/60$, $n = 6$, seven ordinates are available,

$$\therefore \quad S = \int_0^{12/60} v\,dt = \frac{3h}{8}\{(v_0+v_6) + 3(v_1+v_2+v_4+v_5) + 2v_3\}$$

$$\therefore \quad S = \frac{3}{8}\cdot\frac{2}{60}\{(0+0) + 3(20+36+64+42) + 2(60)\} = 7.575$$

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3. Evaluate $\int_0^1 e^{-x^2} dx$ using Simpson's 3/8th rule by dividing $[0,1]$ into 6 equal parts.

Sol: Here $I = \int_0^1 f(x)dx$, $y = f(x) = e^{-x^2}$

$n = 6$ (given), $h = \frac{(1-0)}{6} = \frac{1}{6}$

Note that 6 subdivisions has 7 ordinates.

Construct a table where x varies from 0 to 1 in steps of $1/6$ and tabulate values of y for each 'x', as below:

	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	0	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	1
y	1	0.9726	0.8948	0.7788	0.6412	0.4994	0.3679
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

$$\therefore \quad I = \frac{3h}{8}\{(y_0+y_6) + 3(y_1+y_2+y_4+y_5) + 2y_3\}$$

$$= \frac{3}{8}\cdot\frac{1}{6}\left[(1+0.3679) + 3(0.9726 + 0.8948 + 0.6412 + 0.4994) + 2(0.7788)\right]$$

$$\approx 0.7468$$

4. Evaluate $\int_0^{\pi/2} \sqrt{\sin(x)}\, dx$ using Weddle's rule taking seven ordinates.

Sol: Here $I = \int_0^{\pi/2} \sqrt{\sin(x)}\, dx$, $(n+1) = 7 \quad \therefore n = 6$

$y = f(x) = \sqrt{\sin(x)}$, $h = \frac{(\pi/2 - 0)}{6} = \pi/12$

Seven ordinates corresponds to six subintervals.

	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	0	$\pi/12$	$2\pi/12$	$3\pi/12$	$4\pi/12$	$5\pi/12$	$\pi/2$
y	0	0.5087	0.7071	0.8409	0.9306	0.9828	1
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

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∴ By Weddle's rule:

$$I = \frac{3h}{10}[y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6]$$

$$= \frac{3}{10} \cdot \frac{\pi}{12}[0 + 5(0.5087) + 0.7071 + 6(0.8409) + 0.9306 + 5(0.9828) + 1]$$

$$I = \frac{\pi}{40} \times (15.1406) = 1.18914$$

$$\therefore \int_0^{\pi/2} \sqrt{\sin(x)}\, dx \approx 1.18914$$

Gaussian Quadrature:

Newton-Cotes quadrature uses equally spaced of nodes, whereas Gaussian quadrature selects unequal spacing of nodes and $f(x)$ is approximated by a polynomial of higher order giving better approximation for integral.

Gauss two and three point rule are respectively:

$$\int_{-1}^{1} f(x)\, dx = [f(-1/\sqrt{3}) + f(1/\sqrt{3})]$$

$$\int_{-1}^{1} f(x)\, dx = \left[\frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right)\right]$$

The above integral requires limits to be from -1 to 1, however $\int_a^b f(x)\, dx$ can be transformed to above type by changing the variable.

$$\text{put } x = \left(\frac{b-a}{2}\right)t + \left(\frac{b+a}{2}\right)$$

$$\therefore dx = \left(\frac{b-a}{2}\right) dt$$

this substitution changes the given integral to required form.

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Examples :

1. Evaluate $\int_{-1}^{1} \frac{dx}{(1+x^2)}$ using Gauss three point rule.

Sol : Let $I = \int_{-1}^{1} \frac{dx}{(1+x^2)} = \int_{-1}^{1} f(x)\,dx$

$$f(x) = \frac{1}{(1+x^2)}$$

$$\therefore \quad I = \left[\frac{5}{9} f\left(-\sqrt{\tfrac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\tfrac{3}{5}}\right)\right]$$

$$\therefore \quad f(0) = \frac{1}{(1+0)} = 1, \quad f\left(\sqrt{\tfrac{3}{5}}\right) = 0.625$$

$$f\left(-\sqrt{\tfrac{3}{5}}\right) = 0.625$$

$$\therefore \quad I = \int_{-1}^{1} \frac{dx}{(1+x^2)} \approx 1.58333$$

2. Evaluate $\int_{0}^{1} \frac{dx}{(1+x)}$ using Gauss three point rule, hence obtain an approximate value of $\log_e(2)$.

Sol : Here limits of integration are not -1 to 1

$$\therefore \quad I = \int_{0}^{1} f(x)\,dx, \quad \text{put } x = \left(\frac{1-0}{2}\right)t + \left(\frac{1+0}{2}\right)$$

$$\therefore \quad x = \frac{1}{2}(t+1) \quad \therefore \quad dx = \frac{1}{2}\,dt$$

when $x = 0, \ t = -1, \quad x = 1, \ t = 1$

$$\therefore \quad I = \int_{-1}^{1} \frac{(1/2)\,dt}{\left[1 + \frac{(t+1)}{2}\right]} = \int_{-1}^{1} \frac{dt}{(t+3)}$$

Hence $f(t) = \frac{1}{(t+3)}$

$$f(0) = 1/3 = 0.33333, \quad f\left(\sqrt{\tfrac{3}{5}}\right) = 0.26493$$

$$f\left(-\sqrt{3/5}\right) = 0.44936$$

$$\therefore \quad I = \frac{1}{9}\left[5f\left(-\sqrt{\tfrac{3}{5}}\right) + 8f(0) + 5f\left(\sqrt{\tfrac{3}{5}}\right)\right]$$

$$I = 0.69312, \text{ by integration, } \log(2) \approx 0.69312$$

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Practice examples :

1. Find $f'(1.1)$ and $f'(1.6)$ from the following data :

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
y	10.50	10.75	10.99	11.22	11.45	11.68	11.9

here $y = f(x)$.

$(2.483, 2.117)$

2. A smooth curve passes through the points $(0, 10)$, $(1, 21)$, $(2, 6)$, $(3, 43)$ and $(4, 66)$, find the slope of the curve at $x = 4$

$\left(\text{slope} = \left(\frac{dy}{dx}\right)_{x=4} = 430 \right)$

3. Evaluate $\int_0^2 \frac{dx}{(1+x^3)}$ using Weddle's rule taking seven ordinates.

$(I = 1.0889)$

4. Evaluate $\int_0^{\pi/2} e^{\sin(x)}\, dx$ by dividing $(0, \pi/2)$ is 12 equal parts using Simpson's 1/3rd rule.

5. Evaluate $\int_{0.2}^{1.4} [\sin(x) - \log_e(x) + e^x]\, dx$ by dividing the interval to six parts using Simpson's 3/8th rule and Weddle's rule, Compare the values with the exact solution.

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